

사용한 데이터셋 : 학습 시간, 이전 성적 등 몇가지 조건에 따른 점수 예측
[Student Performance \(Multiple Linear Regression\)](#)

사용한 Features : Hours Studied, Previous Scores, Sleep Hours,
Sample Question Papers Practiced(총 4개)

출력 값 : Performance Index

트레이닝 결과

레이어 수 : 4-64-32-1

에포크 수 : 100회

학습률 : 0.001

데이터 수 : 10000(트레이닝용 8000 : 테스트용 2000)

```
RegressionModel(  
  (fc1): Linear(in_features=4, out_features=64, bias=True)  
  (relu1): ReLU()  
  (fc2): Linear(in_features=64, out_features=32, bias=True)  
  (relu2): ReLU()  
  (fc3): Linear(in_features=32, out_features=1, bias=True)  
)
```

Model moved to: xpu

Loss function MSELoss()

Optimizer: Adam

Learning rate: 0.001

Starting training...

Epoch	[10/100]		Train Loss: 6.4604		Test Loss: 5.7898
Epoch	[20/100]		Train Loss: 4.6290		Test Loss: 4.5509
Epoch	[30/100]		Train Loss: 4.4511		Test Loss: 4.2679
Epoch	[40/100]		Train Loss: 4.3886		Test Loss: 4.3074
Epoch	[50/100]		Train Loss: 4.3289		Test Loss: 4.4703
Epoch	[60/100]		Train Loss: 4.3683		Test Loss: 4.3169
Epoch	[70/100]		Train Loss: 4.2994		Test Loss: 4.3201
Epoch	[80/100]		Train Loss: 4.2978		Test Loss: 4.3230
Epoch	[90/100]		Train Loss: 4.3436		Test Loss: 4.3186
Epoch	[100/100]		Train Loss: 4.2796		Test Loss: 4.3386

Training complete!

#문제점

```
RegressionModel(  
  (fc1): Linear(in_features=4, out_features=64, bias=True)  
  (relu1): ReLU()  
  (fc2): Linear(in_features=64, out_features=32, bias=True)  
  (relu2): ReLU()  
  (fc3): Linear(in_features=32, out_features=1, bias=True)  
)
```

Model moved to: xpu
Loss function: MSELoss()
Optimizer: Adam
Learning rate: 0.001
Starting training...

Epoch	Train Loss	Test Loss
[10/100]	6.4604	5.7898
[20/100]	4.6290	4.5509
[30/100]	4.4511	4.2679
[40/100]	4.3886	4.3074
[50/100]	4.3289	4.4703
[60/100]	4.3683	4.3169
[70/100]	4.2994	4.3664
[80/100]	4.2978	4.3186
[90/100]	4.3436	4.3186
[100/100]	4.2796	4.3386

진동 발생

과적합 발생

Training complete!

#해결방안



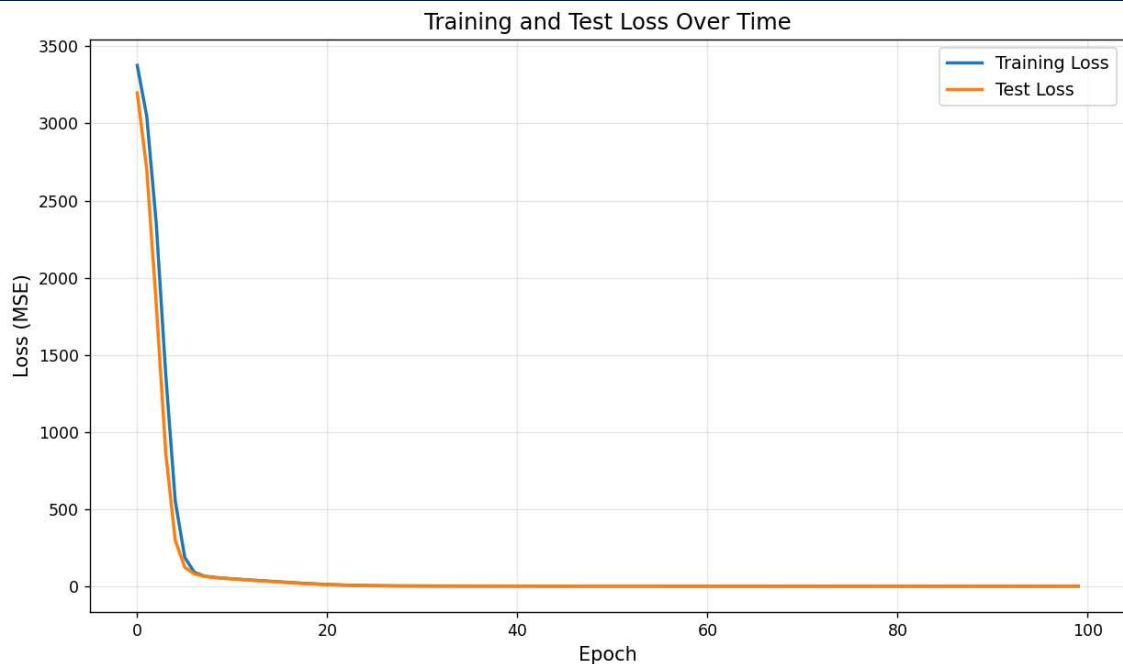
따라서 학습률을 낮춰서 학습한다.

#학습률을 0.0003으로 낮추는 수정 후

```
Model moved to: xpu
Loss function: MSELoss()
Optimizer: Adam
Learning rate: 0.0003
Starting training...
```

```
-----
Epoch [10/100] | Train Loss: 56.3229 | Test Loss: 55.4346
Epoch [20/100] | Train Loss: 18.0409 | Test Loss: 16.8020
Epoch [30/100] | Train Loss: 5.5657 | Test Loss: 5.3393
Epoch [40/100] | Train Loss: 4.8991 | Test Loss: 4.7229
Epoch [50/100] | Train Loss: 4.5729 | Test Loss: 4.4465
Epoch [60/100] | Train Loss: 4.4499 | Test Loss: 4.3370
Epoch [70/100] | Train Loss: 4.3947 | Test Loss: 4.3222
Epoch [80/100] | Train Loss: 4.3921 | Test Loss: 4.3641
Epoch [90/100] | Train Loss: 4.3640 | Test Loss: 4.3424
Epoch [100/100] | Train Loss: 4.3204 | Test Loss: 4.2485
-----
```

Training complete!



학습률을 높임으로써 진동과 과적합문제 둘 다 개선되었다.

실제값 대비 예측그래프 출력

